

Introduction to Contract Theory and Auctions

An introduction to Interdependent Value Auctions

Prof. Alejandro Melo-Ponce



NAZARBAYEV
UNIVERSITY

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On New Year's Eve of A.D. 192, the emperor Commodus was assassinated. This inaugurated the Year of the Five Emperors, A.D. 193. Pertinax was the first to claim the title. The Prætorian guard, which proclaimed him emperor, expected a generous donativum on his ascension, but they were disappointed. On the 28 of march of 193, Pertinax was at the palace when a contingent of three hundred soldiers from the Prætorian Guard rushed the gates, claiming that they had received only half of the promised pay. Pertinax attempted to reason with the insurgents but he was struck down by one of the soldiers.

After this bold move, the Prætorian Guard proceeded to sell off the entire Roman Empire by means of an auction. The winning bid was a promise of 25,000 sesterces per man to the Guard. The winner was the wealthy senator Didius Julianus, who was duly declared emperor. His reign lasted for only sixty five days, after which he suffered from what is perhaps the earliest and most extreme instance of the "winner's curse". He was beheaded.

His last words were "But what evil have I done? Whom have I killed?"

You are about to learn what he did wrong.

Relaxing the Assumption of Private Values

- Private Values: each bidder knows the value of the object to herself.
- We will now allow bidders to only have partial information regarding the value, in the form of a noisy signal.
- Bidders now have information that would, if known to other bidders, affect the value they assign to the object.
- This structure is called **interdependent values**.
- Suppose that Bidder i 's private information is summarized as the realization of a random variable $X_i \in [0, \omega_i]$, which is called his **signal** or type.
- Assume that the value of the object to bidder i , the random variable V_i can be expressed as a function

$$V_i = \nu_i(X_1, X_2, \dots, X_n)$$

where $\nu_i^n : \prod_i [0, \omega_i] \rightarrow \mathbb{R}_+$ is nondecreasing in all its variables and twice continuously differentiable. We will also assume that ν_i is strictly increasing in X_i .

- The specification above assumes that the value is completely determined by the signals, i.e., there is no remaining uncertainty.
 - Suppose that $V_i, i = 1, \dots, n$ denote the n unknown values to the bidders; $X_i, i = 1, \dots, n$ denote the n signals available to the bidders; and S a signal that is only available to the seller. Then define

$$\nu_i(x_1, \dots, x_n) = E[V_i | X_1 = x_1, X_2 = x_2, \dots, X_n = x_n]$$

- In either case, assume that $\nu_i(0, \dots, 0) = 0$ and that $E[V_i] < \infty$.
- We continue to assume that bidders are risk neutral and we will only consider standard auction formats: each bidder maximizes $V_i - p_i$ where p_i is the price paid.
 - We could use a generalization of definition 5.1 to interdependent values to consider other different instances of a sealed-bid auction.
- Private values: $\nu_i(\mathbf{X}) = X_i$.
- An important special case which is at the other end of the spectrum is a **Pure Common Value Auction**, in which all bidders assign the same value:

$$V = \nu(X_1, X_2, \dots, X_n)$$

Other assumptions and remarks

- A special case of the common value auction is when we start by specifying the distribution of V and then, assuming the conditional event $V = v$, we assume that bidder's signals X_i are independently distributed, i.e, **conditional independence** of X_i 's.
- Typically, it is also assumed that $E[X_i|V = v] = v$, i.e., X_i is an **unbiased estimator of the value**.

Things to be careful about:

- The interdependence of values complicates the decision problem of a bidder.
- Since the exact value of the object is unknown and depends on other bidder's signals, an *a priori* estimate may need to be revised as a result of the events taking place **during and after** the auction.
- Reason: events that happen during and after the auction convey **valuable information** about the signals of other bidders.
- One event that is crucial is the announcement of the bidder who has won.

The Winner's Curse

- Prior to the auction, the only information available to $i = 1$ is $X_1 = x$. Based on this, her estimate is $E[V|X_1 = x]$.
- Suppose the object is sold via a FPA and it is announced that 1 is the winner.
- If all bidders are symmetric and follow the same strategy β , what fact does being the winner reveal?
 - That the highest of the other $n - 1$ signals is less than x !
 - As a result:

$$E[V|X_1 = x, Y_1 < x] < E[V|X_1 = x]$$

- As a result, the announcement of winning leads to a decrease in the estimated value, i.e., **bad news!**
- A failure to foresee the above and take it into account is called **the winner's curse**.

The Winner's Curse

- Consider a pure common value auction, in which the bidder's signal is $X_i = V + \epsilon_i$.
- Suppose ϵ_i are i.i.d, and $E[\epsilon_i] = 0$.
- Then $E[X_i|V = v] = v$ and X_i is an unbiased estimator.
- However, the largest of n signals is not unbiased: using Jensen's inequality for convex functions, and with $\max$ being a convex function:

$$E[\max_i X_i | V = v] > \max_i E[X_i | V = v] = v.$$

- A bidder who does not take this into account and bids an amount $\beta(X_i)$ which is close to X_i would, upon winning, pay more than the estimated worth of the object.
- The winner's curse only arises if bidders do not calculate the value of winning correctly and overbid. Rational players playing an equilibrium strategy will not suffer from the winner's curse.

Detour: concave and convex functions

Some facts about convex (concave) functions:

- ① An increasing and convex (concave) function must be continuous (but not necessarily differentiable.)
- ② *Jensen's inequality.* If u is convex (concave), then for any finite collection of numbers $(\alpha_k)_{k=1,\dots,K}$ that add up to 1, $\sum_{k=1}^K \alpha_k u(x_k) \geq (\leq) u\left(\sum_{k=1}^K \alpha_k x_k\right)$. Alternatively, if F is a CDF, if u is convex (concave)

$$\int u(s) dF(s) \geq (\leq) u\left(\int s dF(s)\right).$$

- ③ *Three Strings Lemma:* For any $a < b < c$, we have

$$\frac{u(c) - u(b)}{c - b} \geq (\leq) \frac{u(c) - u(a)}{c - a} \geq (\leq) \frac{u(b) - u(a)}{b - a}.$$

- ④ If u is twice differentiable, then for any $a < c$, $u'(a) \geq (\leq) u'(c)$, and thus $u''(x) \geq (\leq) 0$ for all x .

Non-equivalence of the SPA and the English Auction

- Under private values, we discussed that the SPA and the English Auction were equivalent.
- Why is this no longer the case with interdependent values?
 - In the English auction, active bidders get to know at which prices other bidders have dropped.
 - A SPA makes this information unavailable to the bidders by its very nature.
- Strategic equivalence of both auction formats is recovered if either:
 - 1 There are only two bidders.
 - 2 Values are private.

An Example with Independent Signals: FPA

$V = X_1 + X_2$, X_1, X_2 are i.i.d $U[0, 1]$

FPA: Suppose bidder 2 follows a continuous, strictly increasing strategy $\beta: [0, 1] \rightarrow \mathbb{R}_+$ ($\beta(x) = b$). Bidder 1, after her signal $X_1 = x$ will choose a bid b_1 , to maximize her payoff.

$$\hookrightarrow b_1 = \beta(z)$$

$$u_1(b_1 = \beta(z), \beta(x_2), (X_1, X_2) | X_1 = x) = \int (x + x_2 - \beta(z)) dx_2$$

Bidder 1 wins when $\beta(z) > \beta(x_2)$
 $\Leftrightarrow z > x_2$

Region in bidder 1 wins

$$= \int_0^z (x + x_2 - \beta(z)) dx_2$$

$$= x x_2 + \frac{x_2^2}{2} - \beta(z) \cdot x_2 \Big|_{x_2=0}^{x_2=z}$$

$$= x z + \frac{z^2}{2} - \beta(z) z$$

Differentiate wrt. z : $x + z - \beta'(z) z - \beta(z) = 0 \Rightarrow x + z = \beta'(z) z + \beta(z)$
and take FOC.

$x+z = \beta'(z)z + \beta(z)$, At an equilibrium (symmetric) $x=z \Rightarrow$

$$\Rightarrow 2x = \beta'(x)x + \beta(x) = (\beta(x)x)'$$

bidder 1 with signal $x=0$,

bids $b_0 \geq 0$ (I cannot dismiss $b_0 > 0$)

$$\int_0^x 2s ds = \beta(s) \cdot s \Big|_{s=0}^{s=x} = \beta(x) \cdot x - \cancel{\beta(0) \cdot 0}^{b_0}$$

$$s^2 \Big|_0^x = x^2 = \beta(x) \cdot x \Rightarrow \beta^I(x) = x \quad \text{if } x > 0$$

$$\beta^I(0) = b_0, \quad \text{since } \beta^I \text{ is continuous} \quad b_0 = \lim_{x \rightarrow 0} \beta^I(x) = \lim_{x \rightarrow 0} x = 0$$

$$\beta^I(x) = x, \quad x \in [0, 1]$$

$$\text{Expected Revenues: } E[\beta^I(\max\{X_1, X_2\})]$$

$$= E[\max\{X_1, X_2\}] = 2/3$$

An Example with Independent Signals: SPA

$V = X_1 + X_2$, X_1, X_2 are i.i.d $U[0, 1]$

SPA: All Bidders follow sym monotone increasing strategy β . Bidder 2 receives signal x_2 and bids $\beta(x_2)$. However bidder 1 bids $b_1 = \beta(z)$ in order to maximize her payoff:

$$u_1(\beta(z), \beta(x_2), (x_1, x_2) | x_1 = x_1) = \int_{\beta(z) > \beta(x_2) \Leftrightarrow z > x_2} (x_1 + x_2 - \beta(x_2)) dx_2$$

✓ $\text{pdf } x_2 | x_1 = \text{pdf } x_2 = 1$

$$= \int_0^z (x_1 + x_2 - \beta(x_2)) dx_2$$

$$\frac{\partial u_1}{\partial z} = x_1 + z - \beta(z)$$

At equilibrium $\frac{\partial u_1}{\partial z} = 0$ when $z = x_1 \Rightarrow 2x_1 = \beta^*(x_1)$

$$\begin{aligned} E[X_1 + X_2 | X_1 = x_1, x_1 > X_2] &= E[x_1 + X_2 | x_1 > X_2] = x_1 + E[X_2 | x_1 > X_2] \\ &= x_1 + \frac{1}{x_1} \int_0^{x_1} x_2 dx_2 = \frac{3}{2} x_1 \end{aligned}$$

An Example with Independent Signals: Revenue Comparison

Expected Revenue in the SPA?

$$\begin{aligned} E[\min\{2X_1, 2X_2\}] &= 2 E[\min\{X_1, X_2\}] \quad (\text{Reference for this calculation,} \\ &\quad \text{Lecture 5}) \\ &= 2\left(\frac{1}{3}\right) = \frac{2}{3} \end{aligned}$$

Therefore, we conclude that: $E[FPA] = E[SPA]$

Why?

- $V(X_1, X_2) = X_1 + X_2$,
i.e. the valuation
structure
- Independent signals

An example with Correlated Signals

$V = X_1 + X_2$, X_1, X_2 distributed as $f(x_1, x_2) = \frac{4}{5}(1 + x_1 x_2)$, $(x_1, x_2) \in [0, 1]^2$.

Note that $f(x_2|x_1) = \frac{2(1+x_1 x_2)}{2+x_1}$, $f(x_1|x_2) = \frac{2(1+x_1 x_2)}{2+x_2}$.

FPA: Suppose β : continuous
monotone
eq. strategy bidder $\beta(x_2)$
bidder 1 $b_1 = \beta(z)$ in order

to maximize: $u_1 = E[(V - \beta(z)) I_{\{z > x_2\}}(X_2) | X_1 = x_1]$

$$= E[(x_1 + X_2 - \beta(z)) I_{\{z > x_2\}}(X_2) | X_1 = x_1]$$

$$= \int_0^z (x_1 + x_2 - \beta(z)) f(x_2|x_1) dx_2$$

$$= \int_0^z (x_1 + x_2 - \beta(z)) \frac{2(1 + x_1 x_2)}{2 + x_1} dx_2$$

$$\frac{\partial u_1}{\partial z} (2 + x_1) = 2(x_1 + z)(1 + x_1 z) - \beta'(z) \int_0^z 2(1 + x_1 x_2) dx_2 - \beta(z) 2(1 + x_1 z)$$

$$\frac{\partial u_1}{\partial z} = 2(x_1 + z)(1 + x_1 z) - \beta'(z) \int_0^z 2(1 + x_1 x_2) dx_2 - \beta(z) 2(1 + x_1 z)$$

↑ $(2+x_1)$ The above is equal to zero when $z = x_1$

$$4x_1(1+x_1^2) = \beta'(x_1) \int_0^{x_1} 2(1+x_1 x_2) dx_2 + \beta(x_1) 2(1+x_1^2)$$

$$\Rightarrow 4x_1(1+x_1^2) = \beta'(x_1) [2x_1 + x_1^3] + \beta(x_1) 2(1+x_1^2)$$

$$\Rightarrow \frac{4x_1(1+x_1^2)}{2x_1 + x_1^3} = \beta'(x_1) + \beta(x_1) \frac{2(1+x_1^2)}{2x_1 + x_1^3}$$

$$\beta'(x_1) + \beta(x_1) Q(x_1) = R(x_1)$$

$$\text{where } Q(x_1) = \frac{2(1+x_1^2)}{2x_1 + x_1^3}, \quad R(x_1) = \frac{4x_1(1+x_1^2)}{2x_1 + x_1^3}$$

$$\beta'(x_1) + \beta(x_1) Q(x_1) = R(x_1)$$

$$\text{where } Q(x_1) = \frac{2(1+x_1^2)}{2x_1+x_1^3}, \quad R(x_1) = \frac{4x_1(1+x_1^2)}{2x_1+x_1^3}$$

$$P(x_1) = \exp\left(\int \frac{2(1+s^2)}{2s+s^3} ds\right) = x_1 \sqrt{2+x_1^2}$$

$$\begin{aligned} (\beta(x) P(x))' &= R(x_1) P(x_1) = \left(\frac{4x_1(1+x_1^2)}{2x_1+x_1^3}\right) x_1 \sqrt{2+x_1^2} \\ &= \frac{4x_1(1+x_1^2)}{\sqrt{2+x_1^2}} \end{aligned}$$

$$\beta(0) = 0, \text{ initial condition}$$

$$L \quad x_1 = \sqrt{2} \tan \theta$$

$$\frac{\int_0^{x_1} \frac{4s(1+s^2)}{\sqrt{2+s^2}} ds}{x_1 \sqrt{2+x_1^2}} = \beta^1(x_1) = \frac{4(\sqrt{2} - \sqrt{2+x_1^2} + x_1^2 \sqrt{2+x_1^2})}{3x_1 \sqrt{2+x_1^2}}$$

SPA β - sym, monotone
continuous.

Bidder 2 bids $\beta(x_2)$
Bidder 1 bids $\beta(z) = b_1$

$$u_1(\beta(z), \beta(x_2), (x_1, x_2) | X_1 = x_1) = E[(x_1 + X_2 - \beta(X_2)) I_{\{z > x_2\}} | X_2 | X_1 = x_1]$$

$$= \int_0^z (x_1 + x_2 - \beta(x_2)) f(x_2 | x_1) dx_2$$

$$\frac{\partial u_1}{\partial z} = (x_1 + z - \beta(z)) f(x_2 | x_1) = 0 \quad \text{impose } z = x_1$$

$$\Rightarrow [2x_1 - \beta(x_1)] f(x_2 | x_1) = 0 \Rightarrow \beta(x_1) = 2x_1 //$$

Revenues :

$$\begin{aligned}\text{FPA: } R^1 &= \max \{ \beta^I(X_1), \beta^I(X_2) \} \\ &= \beta^I(\max\{X_1, X_2\}) \\ &= \beta^I(Y_1^{(2)})\end{aligned}$$

$$\begin{aligned}P_r(\max\{X_1, X_2\} \leq y) &= P_r(X_1 \leq y, X_2 \leq y) \\ &= \int_0^y \int_0^y f(x_1, x_2) dx_1 dx_2 \\ &= \frac{1}{5} y(2 + y^2)\end{aligned}$$

$$\xrightarrow{\text{PDF}} \frac{4}{5} y(2 + y^2)$$

$$\therefore E[\beta^I(Y_1^{(2)})] \approx 0.735262$$

$$F_{CV} R^2 = 2 \min \{X_1, X_2\} = 2 Y_{(2)}^{(2)}$$

$$Pr(\min \{X_1, X_2\} \leq y) = 1 - Pr(\min \{X_1, X_2\} > y)$$

$$= 1 - Pr(X_1 > y, X_2 > y)$$

$$= 1 - \int_y^1 \int_y^1 f(x_1, x_2) dx_1 dx_2 = 1 - \frac{1}{5} (-5 + 8y - 2y^2 - y^4)$$

$$PDF \Rightarrow \frac{1}{5} (8 - 4y - 4y^3)$$

$$E[2 \min \{X_1, X_2\}] = 0.7466$$

Math Detour: Integrating factor

- Method to solve linear differential equations that have the form:

$$b'(x) + b(x)Q(x) = R(x)$$

- Notice that in the equation above, the LHS has a slight similarity to the derivative of a product $(bp)' = b'p + bp'$.
- Suppose that we multiply the LHS by a function $P(x)$. Then it becomes

$$b'(x)P(x) + b(x)P(x)Q(x).$$

- If we can choose $P(x)$ such that $P'(x) = P(x)Q(x)$, then the LHS can be transformed into the derivative of the product $b(x)P(x)$:

$$\begin{aligned}(b(x)P(x))' &= b'(x)P(x) + b(x)P'(x) \\ &= b'(x)P(x) + b(x)P(x)Q(x) \\ &= (b'(x) + b(x)Q(x))P(x) \\ &= R(x)P(x)\end{aligned}$$

Math Detour: Integrating factor

- $P(x)$ is called the **integrating factor**.
- To find P such that $P' = PQ$ note that

$$(\ln P)' = \frac{P'}{P} = Q$$

- Thus $P(x) = \exp\left(\int Q(s)ds\right)$.
- For the original differential equation, we have:

$$\int (b(s)P(s))'ds = \int R(s)P(s)ds$$

- If we add a boundary condition, e.g. $b(0) = b_0$ we get

$$b(x)P(x) - b_0P(0) = \int_0^x R(s)P(s)ds \Rightarrow b(x) = \frac{\int_0^x R(s)P(s)ds + b_0P(0)}{P(x)}$$